Intent-bound, Constraint-based Delegation for Agentic AI

Today mostly **payments / open-banking / identity vendors** are turning “intent-bound, constraint-based delegation” into concrete products. The main ones:

1. Agentic payments & verifiable intent

- **Google Cloud / AP2 (Agent Payments Protocol)**
  - Publishes and stewards the **AP2** open protocol for agentic payments (Intent / Cart / Payment Mandates with time/amount constraints).
  - Runs a large ecosystem effort: >60 orgs collaborating, including **Adyen, American Express, Ant International, Coinbase, Etsy, Forter, Intuit, JCB, Mastercard, PayPal, Revolut, Salesforce, ServiceNow, UnionPay, Worldpay**, etc.
- **PayPal**
  - Co-developer and early implementer of **AP2**, focused on embedding mandates into its global payments stack to make *agent-driven* payments verifiable and auditable at scale.
- **Worldline**
  - Implements AP2 as a PSP, with detailed docs on **Intent / Cart / Payment Mandates** and delegated (human-not-present) scenarios for autonomous agents.
- **Mastercard – Verifiable Intent**
  - Introduces a **Verifiable Intent** framework linking identity, intent, and action into a single record for agent-assisted and autonomous transactions, co-developed with Google.
  - Used to decide when existing intent is enough vs when additional human confirmation is required.

2. Consent verification & delegated payment limits

- **AffixIO**
  - Provides **consent verification circuits** for agentic payments; consent records include:
    - Which agents are authorized
    - Spend caps (per txn / per period)
    - Merchant/category restrictions
    - Validity period & revocation

    - Every transaction calls a verification API that returns **eligible / ineligible + proof**, so one consent can safely cover many payments without per-txn prompts.
- **Shuttle Global**
    - Focuses on **consent-based AI payments** and the social-engineering / approval-fatigue problem; pushes architectures where the approval surface is separate from the agent and where prompts are limited and high-value.

3. Open banking & OAuth/OIDC consent for agents

- **Konsentus**
    - Works on **agentic consent in open banking**, modelling delegation via policies that encode explicit delegation, constrained inheritance, and visibility of downstream authority—aligned with emerging OpenID/OAuth extensions for delegation and policy constraints.
- **Curity**
    - Identity and API security vendor pushing **time-limited, transaction-bounded consent** for AI agents on top of OAuth:
        - Granular scopes
        - User-defined **amount and time limits** for agents
        - Re-consent for high-privilege actions.
- **Auth0 / Okta**
    - Driving **async human confirmation** for agent actions using **CIBA** flows (decoupled user authorization with push notifications), which is a key pattern for non-bombing, out-of-band approvals for sensitive agent operations.

4. Standards / ecosystem bodies (non-vendor but influential)

- **AP2 Working Group / ap2-protocol.org** – open protocol community around AP2.
- **NIST NCCoE** – exploring **intent-bound delegation** and AI agent authorization in concept papers (e.g., HAID submissions).
- **OpenID Foundation / open-banking alliances** – extending OAuth/OpenID for delegated, policy-constrained agent authorization.

In practice, **Google + PayPal + Mastercard + the AP2 consortium** are leading on *payment / value-bound* intent; **AffixIO, Konsentus, Curity, Auth0/Okta** are pushing the *policy, consent, and verification* side that makes “one human intent → many bounded agent actions” workable without approval bombing.